

LA ROCHE-SUR-YON AP, France

WMO: 073060

Lat: 46.700N

Lon: 1.379W

Elev: 91

StdP: 100.24

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.1	-2.4	-7.1	2.1	-2.9	-5.0	2.5	-1.5	12.7	10.5	11.3	10.3	3.2	30	0.533

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	30.6	20.3	28.7	19.4	26.6	18.7	21.5	28.2	20.5	26.5	19.5	24.8	3.6	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.4	14.3	23.5	18.5	13.5	22.4	17.7	12.8	21.5	62.8	28.1	59.2	26.5	56.0	24.9	26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.9	8.6	7.6	-6.5	34.4	2.3	1.8	-8.2	35.6	-9.5	36.7	-10.8	37.7	-12.5	38.9		
			-6.9	23.1	2.3	1.1	-8.6	23.9	-9.9	24.5	-11.2	25.2	-12.8	26.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.3	6.0	6.3	8.5	10.7	14.0	17.5	19.3	19.1	16.5	13.5	9.1	6.4
	DBStd	5.84	3.85	3.57	3.10	3.07	3.00	3.20	2.81	2.80	2.92	3.25	3.44	3.95
	HDD10.0	516	130	107	66	27	3	0	0	0	0	9	56	119
	HDD18.3	2358	383	337	304	231	139	53	21	22	69	152	278	369
	CDD10.0	1346	5	4	20	47	126	224	288	282	196	117	28	9
	CDD18.3	144	0	0	0	0	4	28	51	45	16	2	0	0
	CDH23.3	1397	0	0	0	6	54	292	474	424	138	8	0	0
	CDH26.7	429	0	0	0	0	8	87	153	148	31	1	0	0
Wind	WSAvg	3.8	4.2	4.3	4.3	4.0	3.7	3.6	3.5	3.3	3.4	3.4	3.7	4.1
Precipitation	PrecAvg	861	98	71	58	65	58	41	48	51	72	101	104	97
	PrecMax	1236	194	161	152	209	123	98	142	121	280	200	272	173
	PrecMin	579	18	4	4	15	11	6	6	5	2	14	26	16
	PrecStd	159	48	44	38	40	26	21	27	33	60	46	54	46
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.1	15.8	20.4	24.2	27.9	32.5	33.3	34.0	29.9	24.1	17.6	14.5
		MCWB	12.7	11.3	13.2	16.0	19.3	21.2	21.0	21.6	19.2	17.7	14.8	13.0
	2%	DB	12.9	13.3	17.6	21.3	24.7	29.1	30.4	30.5	27.1	21.4	16.0	13.6
		MCWB	11.8	10.5	12.2	15.1	17.7	19.8	20.0	20.5	18.1	16.8	14.4	12.4
	5%	DB	12.1	12.0	15.3	18.7	22.4	26.6	28.2	28.0	24.6	19.6	15.0	12.8
		MCWB	10.9	10.3	11.2	13.2	16.4	18.8	19.3	19.5	17.5	16.1	13.6	11.7
	10%	DB	11.2	11.2	13.5	16.4	20.2	24.3	25.9	25.4	22.4	18.2	14.0	12.0
		MCWB	10.2	9.8	10.5	12.1	15.2	17.7	18.6	18.5	16.5	15.5	12.6	10.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.3	12.2	14.0	17.2	20.2	22.6	22.4	23.5	20.9	18.7	16.1	13.6
		MCDB	13.6	13.9	18.4	22.7	26.2	29.5	30.1	30.9	26.5	21.4	16.6	14.1
	2%	WB	12.1	11.4	12.8	15.4	18.3	20.8	21.0	21.6	19.4	17.6	14.7	12.7
		MCDB	12.7	12.6	16.1	20.4	23.4	27.0	28.0	27.9	24.3	20.1	15.6	13.3
	5%	WB	11.2	10.7	11.9	13.7	16.8	19.5	20.0	20.4	18.4	16.7	13.8	11.9
		MCDB	11.9	11.8	14.0	17.6	21.3	25.3	26.3	26.0	22.6	19.0	14.8	12.7
	10%	WB	10.3	10.0	11.1	12.5	15.7	18.3	19.2	19.3	17.4	15.9	12.7	11.1
		MCDB	11.1	11.1	13.0	15.6	19.5	23.3	24.6	23.6	21.1	17.6	13.9	11.9
Mean Daily Temperature Range	5% DB	MDBR	5.6	6.8	8.4	9.3	9.6	10.4	10.4	10.7	10.3	7.9	6.3	5.8
		MCDBR	5.2	7.4	11.1	12.7	13.4	14.4	14.7	15.1	13.5	9.2	6.4	5.4
		MCWBR	4.4	5.5	6.9	7.5	7.3	6.6	6.1	6.1	6.1	5.3	4.9	4.7
	5% WB	MCDBR	4.9	6.4	8.6	11.5	12.1	12.9	12.4	12.5	11.0	7.7	5.7	5.1
		MCWBR	4.5	5.2	5.8	7.1	6.9	6.5	5.7	5.9	5.7	4.9	4.7	4.8
Clear-Sky Solar Irradiance	taub	0.327	0.342	0.369	0.390	0.393	0.403	0.395	0.384	0.364	0.362	0.337	0.322	
	taud	2.441	2.394	2.320	2.265	2.305	2.312	2.335	2.381	2.416	2.426	2.436	2.422	
	Ebn at Noon	761	823	850	864	872	863	865	861	848	788	738	719	
	Edh at Noon	69	88	109	126	125	125	121	111	98	84	69	63	
All-Sky Solar Radiation	RadAvg	1.16	2.10	3.26	4.68	5.52	6.19	5.95	5.26	4.16	2.46	1.42	1.02	
	RadStd	0.12	0.21	0.36	0.47	0.43	0.52	0.30	0.41	0.28	0.20	0.14	0.12	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (21 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page